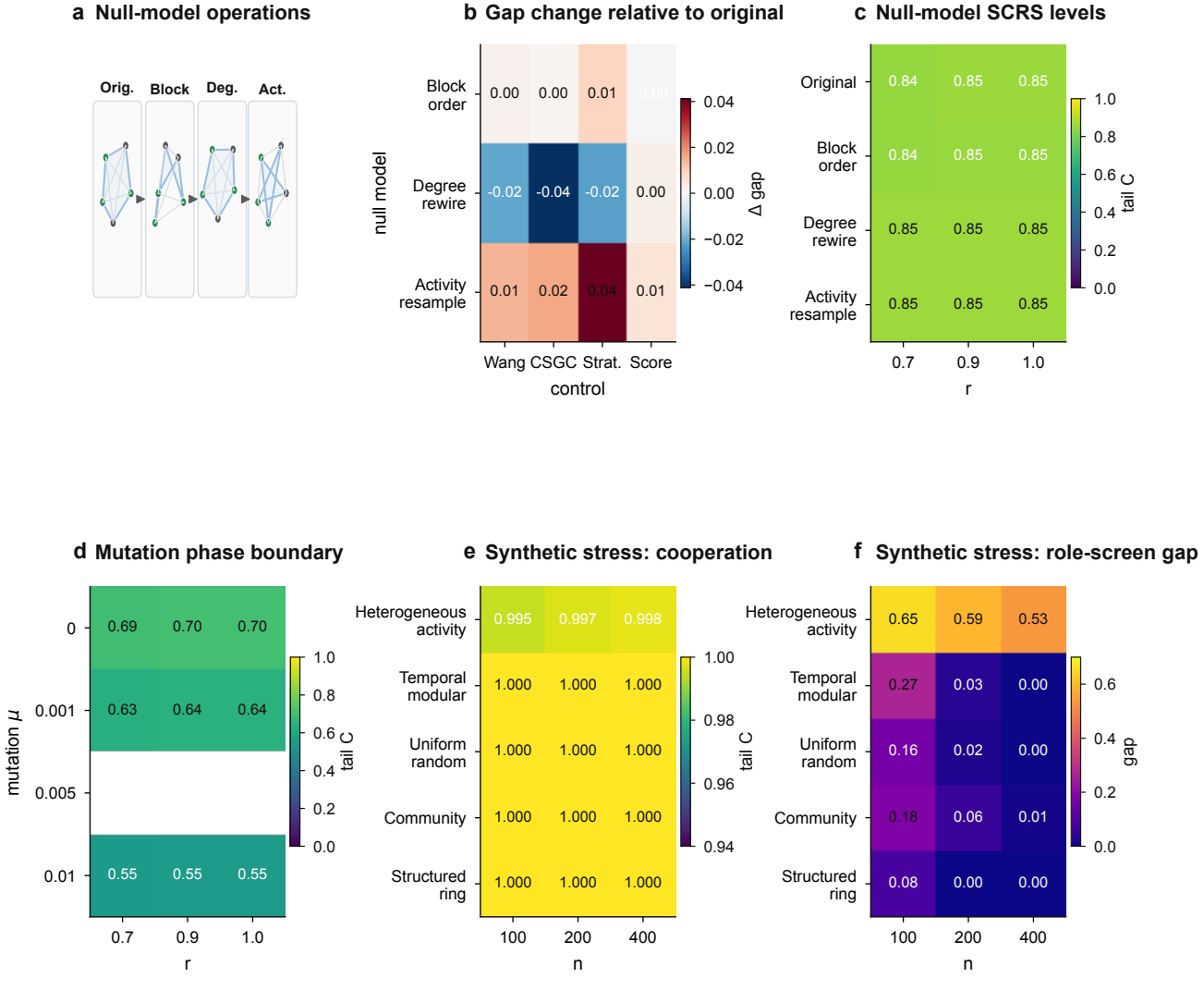

a Null-model operations
Orig.
Block
Deg.
Act.
b Gap change relative to original
null model
Block order
Degree rewire
Activity resample
0.00
0.00
0.01
-0.00
-0.02
-0.04
-0.02
0.00
0.01
0.02
0.04
0.01
Wang
CSGC
Strat.
Score
control
Δ gap
0.04
0.02
0.00
−0.02
−0.04
c Null-model SCRS levels
Original
Block order
Degree rewire
Activity resample
0.84
0.85
0.85
0.84
0.85
0.85
0.85
0.85
0.85
0.85
0.85
0.85
0.7
0.9
1.0
r
tail C
1.0
0.8
0.6
0.4
0.2
0.0
d Mutation phase boundary
mutation μ
0
0.001
0.005
0.01
0.69
0.70
0.70
0.63
0.64
0.64
0.55
0.55
0.55
0.7
0.9
1.0
r
tail C
1.0
0.8
0.6
0.4
0.2
0.0
e Synthetic stress: cooperation
Heterogeneous activity
Temporal modular
Uniform random
Community
Structured ring
0.995
0.997
0.998
1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
100
200
400
n
tail C
1.00
0.98
0.96
0.94
f Synthetic stress: role-screen gap
Heterogeneous activity
Temporal modular
Uniform random
Community
Structured ring
0.65
0.59
0.53
0.27
0.03
0.00
0.16
0.02
0.00
0.18
0.06
0.01
0.08
0.00
0.00
100
200
400
n
gap
0.6
0.4
0.2
0.0